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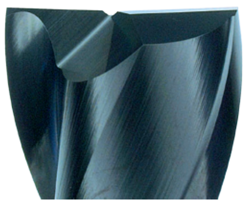
Super Hard Carbon PVD Coating System

HIGH PERFORMANCE COATINGS

Mp€ 500 Ta:€



Tetra amorphous carbon thin film

DO IT YOURSELF
Super Hard Carbon coatings in-house!

MADE IN GERMANY

HIGHLIGHTS

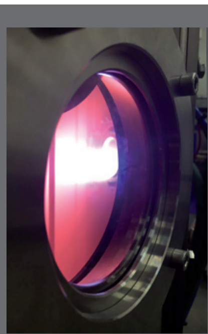
- HPIES High Performance Ion Etching System
- Special designed Ta:€ ARC source
- Smooth coatings
- ~ 80 % sp3 content
- 50 – 60 GPa coating hardness
- Low running costs

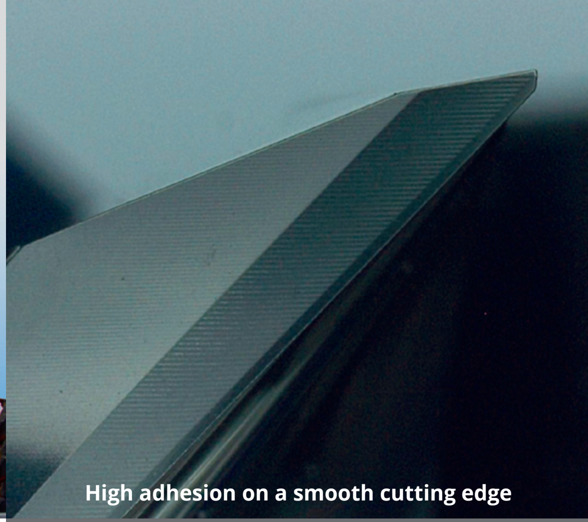
THROUGHPUT

- Useful coating volume: Ø 450 x H 400 mm
- Up to 3 batches / day – Cycle time 6 to 8 hrs
- Qty / batch:
  - end mills Ø10 x 70 mm: 432 pieces
  - drills Ø12 x 150 mm: 288 pieces
  - inserts: 2.400 pieces

CUSTOMER BENEFITS

- Excellent cost per piece ratio by doing Ta:€ coatings in-house (compared to job coating services)
- Enables **new markets for carbide tool makers** by replacing PCD tools through carbide tools + Ta:€ coating in a wide range of applications





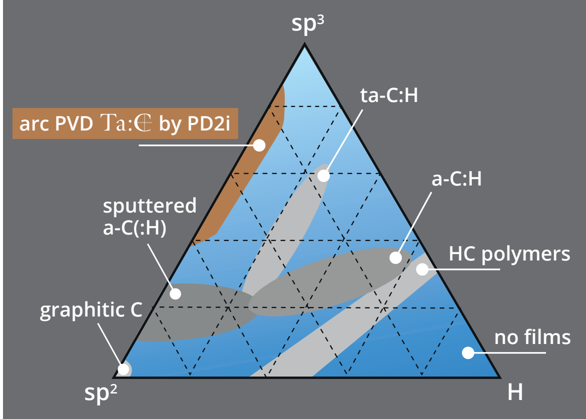
Applications

- Machining
  - Aluminum (all types; soft or high Si containing materials)
  - Copper / Brass
  - PCBs
  - Carbon
  - Plastic composites
  - Noble material (e.g. gold, platinum, titanium) machining
- IC punching
- Precision components
- Functional decorative

Tools

- Routers
- End mills
- Drills
- Punches
- Paper knifes
- Saw blades

The ratio of sp3/sp2 bonds and the hydrogen content in the coating determine the properties of DLC films



Types of DLC Coatings

Characteristics / Properties	
Material	ta-C with ~ 80 % sp3 content
Coating thickness	0,2 to 1,2 µ
Coating hardness	50 - 60 GPa (5.500 - 6.000 HV)
Coefficient of friction	0,1
Max. operating temperatur	500° C / 932° F
Max. deposition temperatur	120° C / 248° F
Color	Rainbow to gray*
Adhesion	Superior through HPIES etching
Biocompatibility	✓
Corrosion resistant	✓

*dep. coating thickness

Advantages

- Reduced sticking effect (e.g. soft Al)
- Improved wear resistance (e.g. Al with high Si content)
- Improved finishing of milling surfaces (e.g. no burs in PCB applications)
- Excellent tribolocial properties

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